

Question number	Answer	Additional guidance	Mark
(a)(i)	An answer that includes the following point - B is the correct answer - A is not the correct answer - C is not the correct answer - D is not the correct answer		(1)

Question number	Answer	Additional guidance	Mark
(a)(ii)	An answer that includes the following point: - B is the correct answer - A is not the correct pathway - C is not the correct pathway - D is not the correct pathway		(1)

Question number	Answer	Additional guidance	Mark
(b)(i)	An answer that includes three of the following points: - heart rate decreases when diving (1) - the longer the whale is diving the lower the heart rate (1) - the difference between maximum heart rate at the surface and minimum heart rate in dive increases as dive duration increases (1)	accept ref to max and min HR	(3)

Question number	Answer	Additional guidance	Mark
(b)(ii)	An answer that includes three of the following points: - relevant physiological change during a dive described (1) {pH/oxygen / carbon dioxide concentrations} in the blood monitored by {chemoreceptors / sensors} (1) - increase in impulses from medulla to SAN (1) - increase impulses from SAN leading to increased heart rate (1)	E.g. blood oxygen decreases / {CO₂/lactate} increases / pH decreases / oxygen debt develops accept increased depolarisations	(3)